

p8106_hw1_sy3352

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```
library(glmnet)
library(caret)
library(tidymodels)
library(corrplot)
library(ggplot2)
library(plotmo)
```

```
training_data <- read.csv("housing_training.csv") |> na.omit()
testing_data <- read.csv("housing_test.csv") |> na.omit()

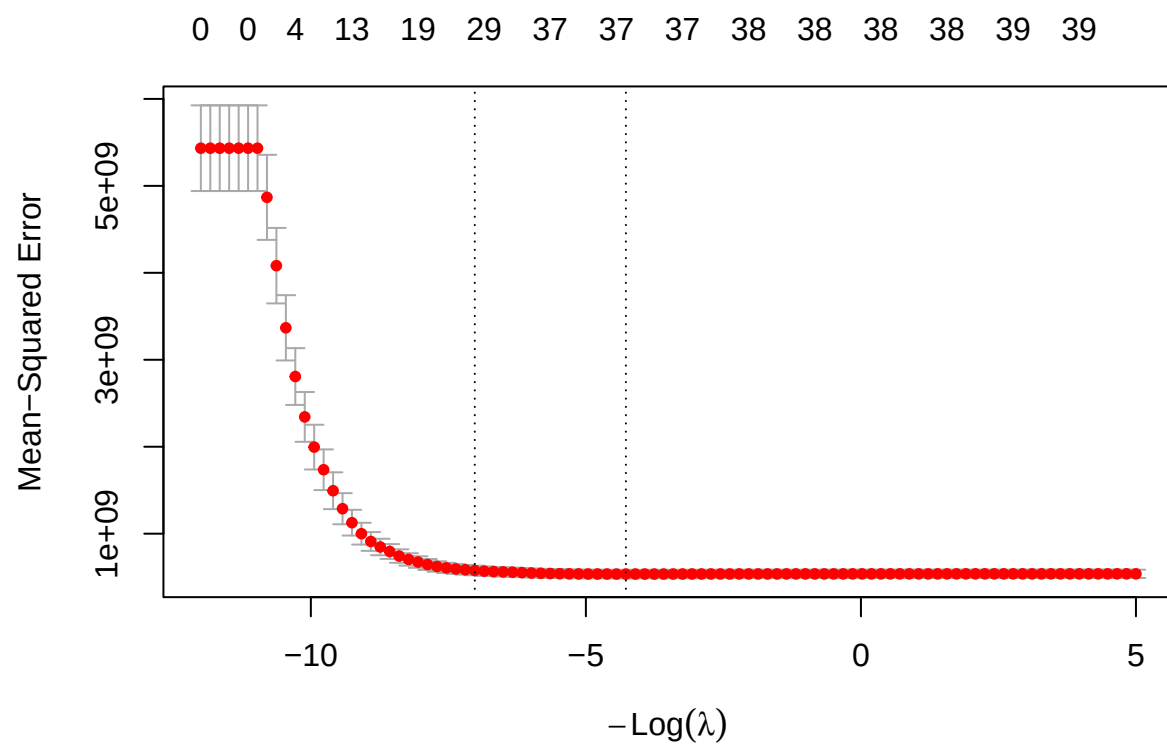
x <- model.matrix(Sale_Price ~ ., training_data)[,-1]
y <- training_data[, "Sale_Price"]

x2 <- model.matrix(Sale_Price ~ ., testing_data)[, -1]
y2 <- testing_data$Sale_Price
```

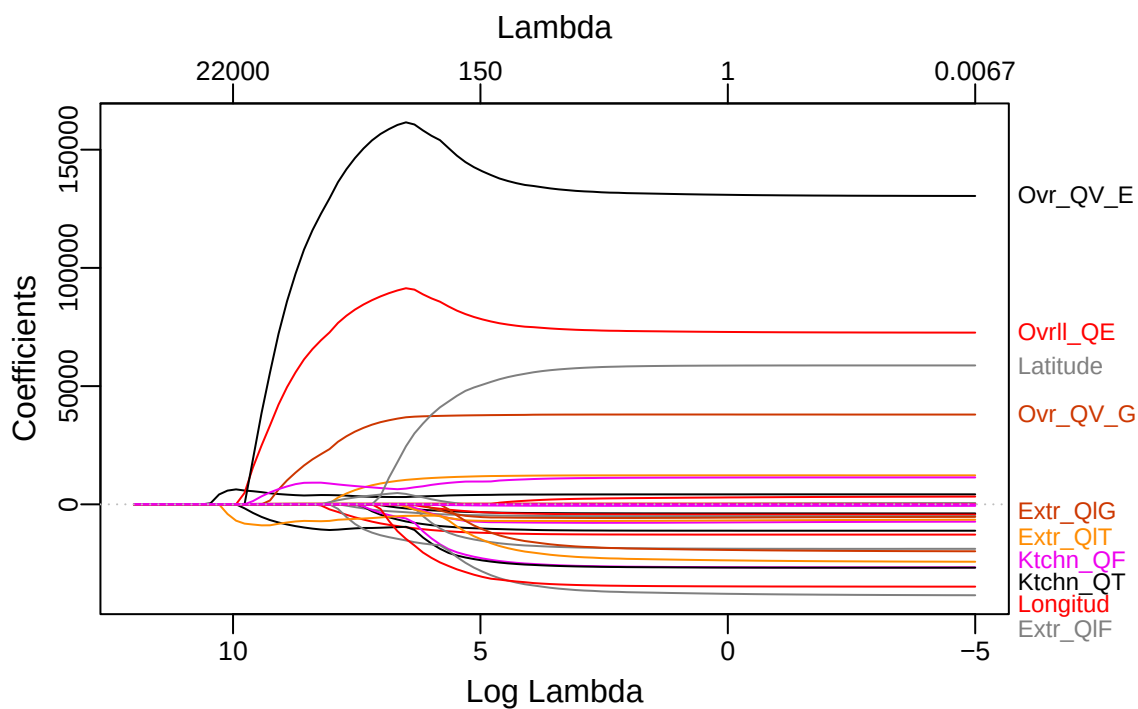
a)

```
set.seed(2026)
cv.lasso <- cv.glmnet(x, y,
                      alpha = 1,
                      lambda = exp(seq(12, -5, length = 100)))

plot(cv.lasso)
```



```
plot_glmnet(cv.lasso$glmnet.fit)
```



```
cv.lasso$lambda.1se
```

```
## [1] 1119.013
```

```
coef_1se <- predict(cv.lasso, s = "lambda.1se", type = "coefficients")
sum(coef_1se != 0) - 1
```

```
## [1] 29
```

```
lasso_pred <- predict(
  cv.lasso, newx = model.matrix(Sale_Price ~ ., testing_data)[,-1],
  s = "lambda.1se", type = "response")

lasso_test_RMSE = sqrt(mean((lasso_pred - testing_data$Sale_Price)^2))
lasso_test_RMSE
```

```
## [1] 20651.86
```

The selected tuning parameter is 1119.0126581. The test error is 2.0651861×10^4 . Using 1SE rule, the number of predictors included is 29.

b)

```

ctrl11 <- trainControl(method = "cv", number = 10)

set.seed(2026)
enet.fit <- train(Sale_Price ~ .,
                  data = training_data,
                  method = "glmnet",
                  tuneGrid = expand.grid(alpha = seq(0, 1, length = 21),
                                         lambda = exp(seq(12, -5, length = 100))),
                  trControl = ctrl11)

```

```

## Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info = trainInfo,
## : There were missing values in resampled performance measures.

```

```
enet.fit$bestTune
```

```

##      alpha  lambda
## 167  0.05 563.0302

```

```

enet.pred <- predict(enet.fit, newdata = testing_data)
enet_test_RMSE = sqrt(mean((enet.pred - testing_data[, "Sale_Price"])^2))
enet_test_RMSE

```

```
## [1] 20947.88
```

The selected tuning parameter is 0.05, 563.030236835952. The test error is 2.094788×10^4 . 1SE is not applicable in elastic net because it has 2 tuning parameters, alpha and lambda. There is no way to apply one standard error of the minimum to 2 parameter. Therefore, 1SE is not applicable here.

c)

```

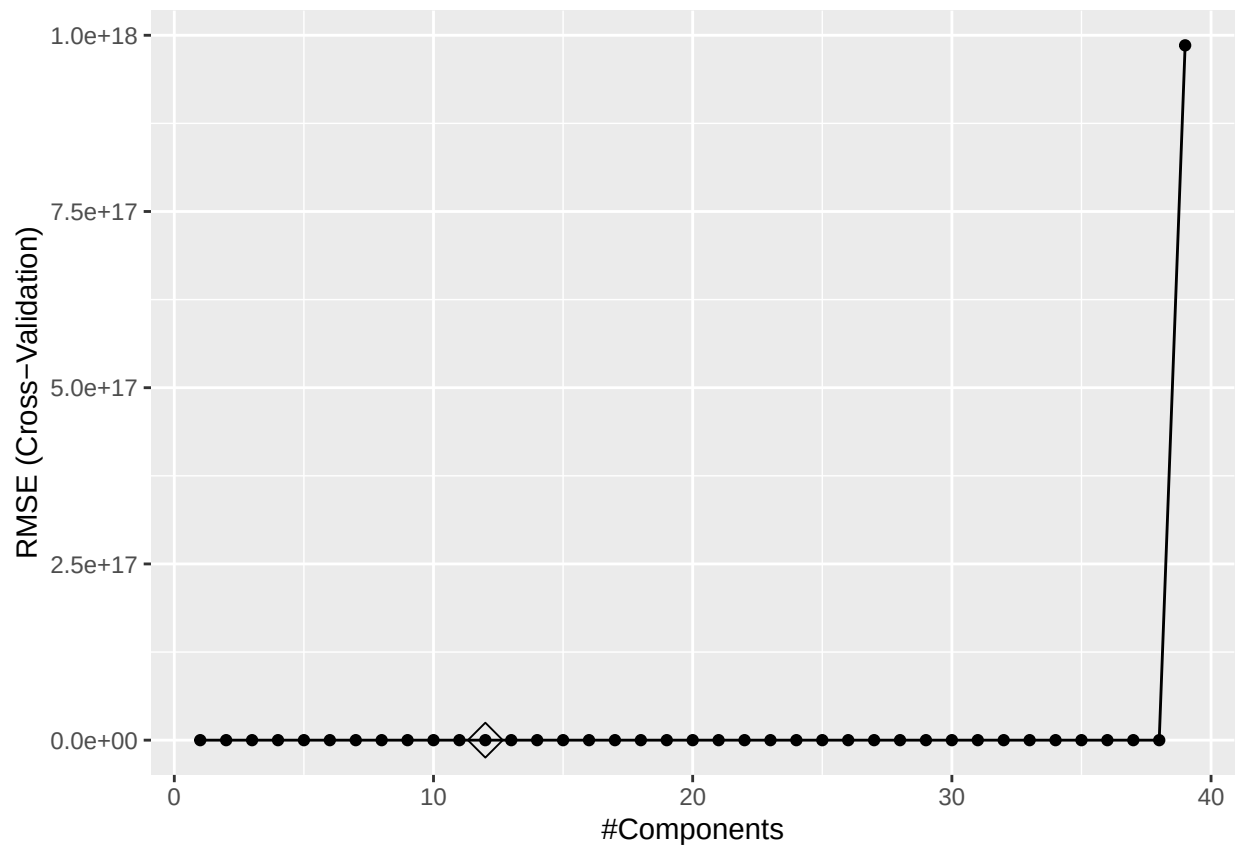
set.seed(2026)
pls_fit <- train(x, y,
                 method = "pls",
                 tuneGrid = data.frame(ncomp = 1:ncol(x)),
                 trControl = ctrl11,
                 preProcess = c("center", "scale"))

predy2_pls2 <- predict(pls_fit, newdata = x2)
pls_test_RMSE = sqrt(mean((y2 - predy2_pls2)^2))
pls_test_RMSE

```

```
## [1] 21204.31
```

```
ggplot(pls_fit, highlight = TRUE)
```



```
pls_fit$bestTune
```

```
##      ncomp
## 12      12
```

The test error is 2.1204309×10^4 . The number of components in the model is 12.

d)

```
lasso_test_RMSE
```

```
## [1] 20651.86
```

```
enet_test_RMSE
```

```
## [1] 20947.88
```

```
pls_test_RMSE
```

```
## [1] 21204.31
```

Among the three tests, the lasso model has the smallest RMSE: 2.0651861×10^4 . Thus it is the best model.